THE REGULAR GRAPHS OF VALENCY AT MOST 3 ON 8 VERTICES

Abstract. Up to isomorphism, the regular graphs of valency at most 3 on 8 vertices are the empty graph, $4K_2$, C_8, C_3+C_5, $2C_4$, $2K_4$, and five connected cubic graphs: the cube, the Wagner graph, $K_{3,3}$ with one vertex replaced by a triangle, K_4 with two vertices replaced by triangles, and two diamonds joined by a matching. This is prerequisite SE4 of `cgss/prerequisites.md`, used in the case E_8 of Remark 4.14 of [CGSS]. We give a proof by cases on a single vertex, explicit isomorphisms, invariants that separate the graphs, a check with SageMath, and the shape of the Lean statements.

1. Statement

All graphs are simple and finite. For a vertex v of a graph X we will write $N(v)$ for its neighbourhood, and $u \sim w$ when u and w are adjacent. We will write $X + Y$ for the disjoint union and mX for m disjoint copies of X.

We will describe each connected cubic graph on 8 vertices as the 8-cycle $0 \sim 1 \sim 2 \sim \cdots \sim 7 \sim 0$ on the vertex set $\{0, \ldots, 7\}$ together with four *chords*, a perfect matching of non-consecutive pairs. We will write $C_8[P]$ for the 8-cycle with the set P of chords. We will say that a cubic graph Y arises from a cubic graph Z by *truncating* a vertex x with neighbours y_1, y_2, y_3 if we delete x and add a triangle $z_1z_2z_3$ with $z_i \sim y_i$.

Definition 1.1. We will use the following cubic graphs on $\{0, \ldots, 7\}$.

(1) The *cube* $Q_3 = C_8[\{03, 16, 25, 47\}]$.
(2) The *Wagner graph* (the Möbius ladder on 8 vertices) $M_8 = C_8[\{04, 15, 26, 37\}]$.
(3) $Y_1 = C_8[\{02, 15, 36, 47\}]$, which is $K_{3,3}$ with one vertex truncated.
(4) $Y_2 = C_8[\{02, 15, 37, 46\}]$, which is K_4 with two vertices truncated.
(5) $Y_4 = C_8[\{02, 13, 46, 57\}]$, which is two copies of the diamond $K_4 - e$ with the two vertices of valency 2 of the one joined by a matching to those of the other.

The subscript of Y_t is its number of triangles. We will also use $2K_4$, with the parts $\{0, 1, 2, 3\}$ and $\{4, 5, 6, 7\}$.

The descriptions by truncation and by diamonds are not needed below; they are checked in Section 5, and we will prove everything from the chord descriptions.

Theorem 1.2. *Let X be a k-regular graph on 8 vertices, with $k \leq 3$. Then X is isomorphic to exactly one of the following graphs:*

(1) *for $k = 0$, the empty graph $\overline{K_8}$;*
(2) *for $k = 1$, $4K_2$;*
(3) *for $k = 2$, C_8, $C_3 + C_5$ or $2C_4$; and*
(4) *for $k = 3$, $2K_4$, Q_3, M_8, Y_1, Y_2 or Y_4.*

Since the complement of a k-regular graph on 8 vertices is $(7 - k)$-regular, we obtain that up to isomorphism and complementation there are exactly 11 regular graphs on 8 vertices; these are the 11 graphs Y of step $\langle 1 \rangle 10$ of Remark 4.14, where $T(8)$ is switched with respect to the edge set of Y (computation C22).

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Each graph of the list is clearly k-regular on 8 vertices, so Theorem 1.2 has two halves: every k-regular graph is isomorphic to one on the list (Sections 2 and 3), and no two graphs on the list are isomorphic (Section 4).

2. Valency at most 2

For $k = 0$ there is nothing to prove. For $k = 1$ every vertex has exactly one neighbour, so the edges of X form a perfect matching $\{u_1w_1, \dots, u_4w_4\}$, and $u_i \mapsto 2i - 2$, $w_i \mapsto 2i - 1$ is an isomorphism onto $4K_2 = \{01, 23, 45, 67\}$.

Lemma 2.1. *A connected 2-regular graph X on m vertices is isomorphic to C_m, and $m \geq 3$.*

Proof. Choose a vertex v_0 and a neighbour v_1 of v_0, and for $i \geq 1$ let v_{i+1} be the neighbour of v_i other than v_{i-1}. Let r be the least index such that $v_r \in \{v_0, \dots, v_{r-1}\}$, say $v_r = v_j$. If $j \geq 1$, then v_j is adjacent to v_{j-1}, v_{j+1} and v_{r-1}; these are distinct, since $r - 1 \neq j \pm 1$ (for $r - 1 = j + 1$ we would have $v_r = v_j = v_{r-2}$, which v_r is chosen to avoid), and this contradicts 2-regularity. Thus $j = 0$, and $v_0, \dots, v_{r-1}$ are distinct and form a cycle, with $r \geq 3$ since X is simple. Every vertex on this cycle has both of its neighbours on it, so the cycle is a connected component of X, and since X is connected, it is all of X. Thus $r = m$, and $v_i \mapsto i$ is an isomorphism onto C_m. □

Let X be 2-regular on 8 vertices. Each connected component of X is 2-regular, and so is a cycle of length at least 3 by Lemma 2.1. The lengths form a partition of 8 into parts of size at least 3; these are 8, $5 + 3$ and $4 + 4$, since three parts would need at least 9 vertices. Thus X is isomorphic to C_8, $C_3 + C_5$ or $2C_4$.

3. Valency 3

Let X be a cubic graph on 8 vertices. For a vertex v, we will write $R(v)$ for the set of the four vertices other than v that are not adjacent to v, and $e(v)$ for the number of edges with both ends in $N(v)$; equivalently, $e(v)$ is the number of triangles through v. For disjoint sets S and T of vertices, $e(S)$ is the number of edges inside S and $e(S, T)$ the number of edges between S and T.

Lemma 3.1. *Let X be a cubic graph on 8 vertices and v a vertex of X. Then*

$$e(N(v), R(v)) = 6 - 2e(v), \qquad e(R(v)) = 3 + e(v).$$

Each vertex of $N(v)$ has exactly $2 -$ (its valency in $N(v)$) neighbours in $R(v)$, and each vertex of $R(v)$ has exactly $3 -$ (its valency in $R(v)$) neighbours in $N(v)$.

Proof. The last sentence holds since v is adjacent to every vertex of $N(v)$ and to no vertex of $R(v)$. Summing the valencies over $N(v)$ gives $9 = 3 + 2e(v) + e(N(v), R(v))$, and summing them over $R(v)$ gives $12 = 2e(R(v)) + e(N(v), R(v))$; the result follows. □

We will also use the graphs on 4 vertices with a given number of edges, which a reader can list by hand, and a kernel can list by deciding over the 64 graphs on $\{1, 2, 3, 4\}$.

Lemma 3.2. *A graph Z on 4 vertices with 6, 5 or 4 edges is K_4, $K_4 - e$, or one of C_4 and the* paw *(a triangle with a pendant edge), respectively. A triangle-free graph on 4 vertices with 3 edges is $K_{1,3}$ or the path P_4.*

Proof. With 4 edges, the complement has 2 edges, which are disjoint (and $Z = C_4$) or share a vertex (and Z is the paw). With 3 edges and no triangle: if a vertex has valency 3, then $Z = K_{1,3}$. Otherwise every vertex has valency at most 2, so Z is a disjoint union of paths and cycles; a cycle would be a triangle or would have 4 edges, so Z is a forest with 4 vertices and 3 edges, and so a tree of maximum valency 2, which is P_4. □

Let $m = \max_v e(v) \in \{0,1,2,3\}$, and choose a vertex v with $e(v) = m$. Write $R = R(v)$. By Lemma 3.1, $e(R) = 3 + m$. In each case below we will label $N(v) = \{x_1, x_2, x_3\}$ and $R = \{w_1, w_2, w_3, w_4\}$ and determine all edges; the conclusion is a Hamiltonian cycle $y_0y_1 \cdots y_7$ of X for which $y_i \mapsto i$ is an isomorphism onto the model, which the reader can check by comparing the four remaining edges with the chords.

Case $m = 3$**.** Then $N(v)$ is a triangle, so $\{v\} \cup N(v)$ spans K_4 and $e(N(v), R) = 0$. Thus $e(R) = 6$ and R spans K_4, and $X \cong 2K_4$ (send v, x_1, x_2, x_3 to $0, 1, 2, 3$ and $w_1, \ldots, w_4$ to $4, \ldots, 7$).

Case $m = 2$**.** The two edges in $N(v)$ share a vertex; label them x_1x_2 and x_2x_3, so $x_1 \not\sim x_3$. Since $e(R) = 5$, R spans $K_4 - e$ (Lemma 3.2); label it so that $w_1 \not\sim w_2$. By Lemma 3.1, x_2 has no neighbour in R, x_1 and x_3 have one each, w_3 and w_4 have no neighbour in $N(v)$, and w_1 and w_2 have one each. Thus the two edges between $N(v)$ and R form a matching of $\{x_1, x_3\}$ with $\{w_1, w_2\}$, and after exchanging w_1 and w_2 we may assume that $x_1 \sim w_1$ and $x_3 \sim w_2$. All edges are now determined, and $x_3\, v\, x_2\, x_1\, w_1\, w_3\, w_4\, w_2$ is a Hamiltonian cycle whose remaining edges vx_1, x_2x_3, w_1w_4, w_3w_2 are the chords 13, 20, 46, 57. Thus $X \cong Y_4$.

Case $m = 1$**.** Here every vertex u satisfies $e(u) \leq 1$. Label $N(v)$ so that $x_1 \sim x_2$ is its edge. By Lemma 3.1, x_1 and x_2 have one neighbour in R each, and x_3 has two. Since $e(R) = 4$, R spans C_4 or the paw (Lemma 3.2).

The paw. Label R so that $w_1w_2w_3$ is the triangle and $w_3 \sim w_4$. By Lemma 3.1, w_3 has no neighbour in $N(v)$, w_1 and w_2 have one each, and w_4 has two. If $x_3 \not\sim w_4$, then $w_4 \sim x_1$ and $w_4 \sim x_2$, and $N(x_1) = \{v, x_2, w_4\}$ contains the edges vx_2 and x_2w_4, so $e(x_1) = 2$, a contradiction. Thus $x_3 \sim w_4$, and after exchanging x_1 and x_2 we may assume that the other neighbour of w_4 in $N(v)$ is x_1. The vertex x_2 and the second R-neighbour of x_3 now lie in $\{w_1, w_2\}$, each of which has exactly one neighbour in $N(v)$; so after exchanging w_1 and w_2 we may assume that $x_2 \sim w_2$ and $x_3 \sim w_1$. Then $v\, x_2\, x_1\, w_4\, w_3\, w_2\, w_1\, x_3$ is a Hamiltonian cycle whose remaining edges vx_1, x_2w_2, w_4x_3, w_3w_1 are the chords 02, 15, 37, 46. Thus $X \cong Y_2$.

The 4-cycle. Label R as the cycle $w_1w_2w_3w_4$. By Lemma 3.1, every w_i has exactly one neighbour in $N(v)$. The two neighbours of x_3 in R are adjacent or opposite on this cycle.

If they are adjacent, then after a rotation of the cycle they are w_1 and w_2. Thus x_1 and x_2 are matched with w_3 and w_4, and after the reflection of the cycle that exchanges w_1 with w_2 and w_3 with w_4 we may assume that $x_1 \sim w_4$ and $x_2 \sim w_3$. Then $x_2\, v\, x_1\, w_4\, w_1\, x_3\, w_2\, w_3$ is a Hamiltonian cycle whose remaining edges x_2x_1, vx_3, w_4w_3, w_1w_2 are the chords 02, 15, 37, 46. Thus $X \cong Y_2$.

If they are opposite, then after a rotation they are w_1 and w_3. Thus x_1 and x_2 are matched with w_2 and w_4, and after exchanging x_1 and x_2 we may assume that $x_1 \sim w_2$ and $x_2 \sim w_4$. Then $x_1\, v\, x_2\, w_4\, w_1\, x_3\, w_3\, w_2$ is a Hamiltonian cycle whose remaining edges x_1x_2, vx_3, w_4w_3, w_1w_2 are the chords 02, 15, 36, 47. Thus $X \cong Y_1$.

Case $m = 0$**.** Then X has no triangle. By Lemma 3.1, every x_i has two neighbours in R, and $e(R) = 3$, so R spans $K_{1,3}$ or P_4 (Lemma 3.2).

The star. Label R so that w_4 is the centre. By Lemma 3.1, w_4 has no neighbour in $N(v)$ and w_1, w_2, w_3 have two each. Thus the edges between $\{x_1, x_2, x_3\}$ and $\{w_1, w_2, w_3\}$ form a 2-regular bipartite graph, whose bipartite complement is a perfect matching; after permuting w_1, w_2, w_3 we may assume that $x_i \not\sim w_i$ and $x_i \sim w_j$ for $i \neq j$. Then $v\, x_1\, w_2\, x_3\, w_1\, w_4\, w_3\, x_2$ is a Hamiltonian cycle whose remaining edges vx_3, x_1w_3, w_2w_4, w_1x_2 are the chords 03, 16, 25, 47. Thus $X \cong Q_3$.

The path. Label R as the path $w_1w_2w_3w_4$. By Lemma 3.1, w_1 and w_4 have two neighbours in $N(v)$, and w_2 and w_3 have one. Since $w_1 \sim w_2$ and X has no triangle, w_1 and w_2 have no common neighbour, so their neighbours in $N(v)$ partition $N(v)$; label $N(v)$ so that $w_1 \sim x_1, x_2$ and $w_2 \sim x_3$. The second neighbour of x_3 in R is not w_1, and is not w_3 (since $x_3w_2w_3$ would be a triangle), so $x_3 \sim w_4$. By the same argument for $w_3 \sim w_4$, the neighbours of w_3 and w_4 in $N(v)$

partition $N(v)$; after exchanging x_1 and x_2 we may assume that $w_4 \sim x_1$, and then $w_3 \sim x_2$. Then $v\,x_3\,w_2\,w_1\,x_1\,w_4\,w_3\,x_2$ is a Hamiltonian cycle whose remaining edges vx_1, x_3w_4, w_2w_3, w_1x_2 are the chords 04, 15, 26, 37. Thus $X \cong M_8$.

This completes the proof that every cubic graph on 8 vertices is isomorphic to one of $2K_4$, Q_3, M_8, Y_1, Y_2 and Y_4. We note that the graph Y_2 arises twice, from different vertices; the case split is on a vertex, not on the graph, and the proof does not need to know which case a given graph falls in. (The computation of Section 5 shows that Q_3 and M_8 have $m = 0$, Y_1 and Y_2 have $m = 1$, Y_4 has $m = 2$, and $2K_4$ has $m = 3$.)

4. Invariants

We will separate the graphs by two numbers that a kernel computation can evaluate on $\{0, \dots, 7\}$ by `decide`: the number $t(X)$ of triangles, and the number $q(X)$ of 4-cycles (subgraphs isomorphic to C_4). If $c(u, w) = |N(u) \cap N(w)|$, then

$$q(X) = \frac{1}{2} \sum_{\{u,w\}} \binom{c(u,w)}{2},$$

where the sum runs over the unordered pairs of distinct vertices, since a 4-cycle $u\,x\,w\,y$ is counted once at each of its two diagonals $\{u, w\}$ and $\{x, y\}$. The valency separates graphs of different valency. Within a valency, the pairs (t, q) are distinct:

$k = 2$	C_8	$C_3 + C_5$	$2C_4$			
(t, q)	$(0, 0)$	$(1, 0)$	$(0, 2)$			
$k = 3$	$2K_4$	Q_3	M_8	Y_1	Y_2	Y_4
(t, q)	$(8, 6)$	$(0, 6)$	$(0, 4)$	$(1, 3)$	$(2, 2)$	$(4, 2)$

These are easy to see by hand from the chord descriptions. For instance, in M_8 the 4-cycles are $i\,(i{+}1)\,(i{+}5)\,(i{+}4)$ for $i = 0, 1, 2, 3$, and M_8 has no triangle since no chord joins two vertices at distance 2 on the 8-cycle. Further invariants, which we do not need, are recorded in Section 5: Q_3 is the only bipartite cubic graph on the list, $2K_4$ the only disconnected one, and the automorphism groups of $2K_4$, Q_3, M_8, Y_1, Y_2, Y_4 have orders 1152, 48, 16, 12, 4 and 16.

5. The computation

The script `cgss/proofs/reg8/checks/reg8_check.sage`, run with SageMath 10, does the following.

(1) It builds the graphs of Definition 1.1 and checks that Q_3 is SageMath's `graphs.CubeGraph(3)`, that M_8 is the 8-cycle with the four long diagonals, that Y_1 is $K_{3,3}$ with one vertex truncated, that Y_2 is K_4 with two vertices truncated, and that Y_4 is the diamond graph of the case $m = 2$.

(2) For $k = 0, 1, 2, 3$ it lists the k-regular graphs on 8 vertices with `nauty_geng("8 -dk -Dk")`; it obtains 1, 1, 3 and 6 graphs, and checks that each is isomorphic to exactly one graph of Theorem 1.2, and conversely.

(3) It computes t, q, connectivity, bipartiteness, girth, diameter and the order of the automorphism group of each graph, and checks that the pairs (t, q) are distinct within each valency.

(4) For each cubic graph it computes $m = \max_v e(v)$ and the shapes of $R(v)$ for the vertices with $e(v) = m$, which agree with Section 3.

(5) It checks the six explicit isomorphisms of Section 3: each case graph, built from the edges determined there, is cubic, and the stated Hamiltonian cycle maps it onto the model.

All checks pass. We note that the computation is an independent check and not a part of the proof, which is complete without it.

6. Lean

We will follow `cgss/notation.md` §10: the namespace `SimpleGraph`, lowerCamelCase for definitions, snake_case theorems naming their conclusion, and `nonempty_iso` for existence of an isomorphism. This section gives the statement shapes that we wrote before the formalization; the Lean formalization, in `LineGraphs/Seidel/`, follows them, and `cgss/notation.md` (§10, the row SE4) gives the final names. Mathlib (at commit `082e2d37` of 4 September 2026, the version pinned in `lake-manifest.json`) has `cycleGraph`, `circulantGraph`, the disjoint sum `SimpleGraph.sum`, `IsRegularOfDegree`, `IsCycles` and Tutte's theorem, but no cube, Wagner or truncated graphs, no statement that a connected 2-regular graph is a cycle, and no Petersen matching theorem. So all the models below are new, and we define them on `Fin 8` to keep adjacency decidable:

graph	Lean
$4K_2$, $C_3 + C_5$, $2C_4$, $2K_4$	`SimpleGraph.fourK2`, `cycleThreeFive`, `twoC4`, `twoK4`
$C_8[P]$	`SimpleGraph.cycleWithChords (P : Finset (Sym2 (Fin 8)))`, the supremum of `cycleGraph 8` and `fromEdgeSet P`
Q_3	`SimpleGraph.cubeGraph`
M_8	`SimpleGraph.wagnerGraph`
Y_1	`SimpleGraph.truncatedK33Graph`
Y_2	`SimpleGraph.twiceTruncatedK4Graph`
Y_4	`SimpleGraph.twoDiamondsGraph`
$q(X)$	`SimpleGraph.numFourCycles`

The docstrings record the chords and the other descriptions. We will state the theorems for `X : SimpleGraph V` with `[Fintype V] [DecidableEq V] [DecidableRel X.Adj]` and `hV : Fintype.card V = 8`. They are

```
theorem SimpleGraph.IsRegularOfDegree.eq_bot_of_zero
    (h : X.IsRegularOfDegree 0) : X = bot
theorem SimpleGraph.IsRegularOfDegree.nonempty_iso_fourK2
    (hV : Fintype.card V = 8) (h : X.IsRegularOfDegree 1) :
    Nonempty (Iso X fourK2)
theorem SimpleGraph.IsRegularOfDegree.nonempty_iso_cycle
    (hc : X.Connected) (h : X.IsRegularOfDegree 2) :
    Nonempty (Iso X (cycleGraph (Fintype.card V)))
```

(with `bot` standing for Mathlib's bottom graph, and the last one, Lemma 2.1, for any finite V), together with the following.

- `SimpleGraph.IsRegularOfDegree.nonempty_iso_of_two_of_card_eq_eight`, with conclusion the disjunction of `Nonempty (Iso X (cycleGraph 8))`, `Nonempty (Iso X cycleThreeFive)` and `Nonempty (Iso X twoC4)`;
- `SimpleGraph.IsRegularOfDegree.nonempty_iso_of_three_of_card_eq_eight`, with conclusion the disjunction over `twoK4`, `cubeGraph`, `wagnerGraph`, `truncatedK33Graph`, `twiceTruncatedK4Graph` and `twoDiamondsGraph`;
- the helpers `SimpleGraph.IsRegularOfDegree.card_edges_neighborFinset_compl` and `card_edges_compl_neighborFinset` for Lemma 3.1, and the four-vertex list of Lemma 3.2 by `decide` on `SimpleGraph (Fin 4)`;

- the invariants: `(cubeGraph.cliqueFinset 3).card = 0`, `cubeGraph.numFourCycles = 6`, and so on for the table of Section 4, each by `decide`; and from these `SimpleGraph.cubic_eight_pairwise_not_iso` and `two_regular_eight_pairwise_not_iso`, via the invariance of both counts under isomorphism (`SimpleGraph.Iso`).

Why this route. We chose the case split on a vertex v maximizing $e(v)$ because every step is local: one counting lemma, a list of graphs on 4 vertices, and at most two exchanges of labels per case, so that a Lean proof chooses v, names $x_1, x_2, x_3, w_1, \ldots, w_4$ by `Finset` choices, and ends by writing down the Hamiltonian cycle as an explicit `Fin 8`-valued bijection whose adjacency the kernel checks by `decide`. The alternatives are longer. A route through a perfect matching needs Petersen's theorem, which Mathlib does not have (it has Tutte's theorem, from which Petersen's theorem is a further argument about bridgeless graphs), and after removing the matching it leaves C_8, $C_3 + C_5$ or $2C_4$ together with a choice of matching, which is more cases. The complement is 4-regular with 16 edges, and its local structure is richer. A brute-force `decide` over graphs on `Fin 8` is out of reach: even after fixing $N(0) = \{1, 2, 3\}$, there remain 2^{21} edge sets on the other pairs.

References

[1] P. J. Cameron, J.-M. Goethals, J. J. Seidel and E. E. Shult, Line graphs, root systems, and elliptic geometry, *J. Algebra* **43** (1976), 305–327.